



Vizhinjam International Seaport Ltd. (VISL)

In December 2013, Vizhinjam International Seaport Ltd (VISL), a Special Purpose Vehicle (SPV) of the Government of the state of Kerala, invited International Competitive Bids (ICB) for selection of Public Private Partnership (PPP) Concessionaire for the proposed Vizhinjam Port. This was the culmination of over a decade of effort to develop an international transshipment port at Vizhinjam in Kerala.

Adani Ports and SEZ Ltd., a subsidiary of Adani Enterprises Limited, the flagship company of the Adani Group, one of India's largest private developers and operators of ports and related infrastructure, was contemplating a bid for the project. While the project was consistent with its strategy of a "pan India presence" it was subject to significant political, construction and market risks. The extent to which the risks could be mitigated through the concession agreement and financing arrangements, including government viability gap funding, would determine the viability of the project from Adani's perspective.

VIZHINJAM INTERNATIONAL SEAPORT DEVELOPMENT OVERVIEW

The Vizhinjam Port location was near the southern tip of India, some 200 km south of Cochin and around 400 km over the sea to Colombo and some 25 km south of the city of Trivandrum. Kovalam beach, an important tourist centre, was located about 2 km towards the North of Vizhinjam village. The Vizhinjam harbor, which had been developed in a natural bay protected by breakwaters, was one of the busiest fishing harbour in the Trivandrum area.

The port location was selected to tap the potential of development of a deep-water international container transshipment port that could handle the largest container vessels serving the East-West shipping route in the Indian Subcontinent region. **Exhibit 1** provides a stylized map of the location. Container transshipment was needed in order to transfer the containers from smaller feeder vessels to larger mainline vessels transiting along the East-West corridor and vice-versa. Container vessels larger than 10,000 TEU¹ provided cost savings as compared to feeder vessels of sizes less than 4,000 TEU. The natural water depth available at Vizhinjam port was more than at any competing Indian port and comparable to competing international ports. The greenfield location provided ample space for the development of world class infrastructure for efficient services as well as for future expansion. With these advantages and planned facilities to handle the biggest container vessels (up to 18,000 TEUs), Vizhinjam would be able to attract mainline vessels to call directly instead of transshipment at other ports outside of the country.

¹ The **twenty-foot equivalent unit (TEU or teu)** is a unit of cargo capacity often used to describe the capacity of container ships and container terminals. It is based on the volume of a 20-foot-long intermodal carrier, a standard-sized metal box which can be easily transferred between different modes of transportation, such as ships, trains and trucks. The container is defined by its length though there is a lack of standardisation in regard to height.

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Vizhinjam port would face direct competition from the port of Colombo and the International Container Transshipment Terminal (ICTT) at Kochi for the Indian sub-continent (ISC) regional traffic, as well as from major transshipment hubs outside the ISC region such as Aden in the west and Singapore in the east.

Vizhinjam Port was expected to attract low volume of “gateway” traffic, i.e. port traffic originating in or destined for Kerala. This was primarily because of a few industries in the immediate hinterland of the port. In any case, Vizhinjam port would be competing with Cochin and Tuticorin for its gateway containerized cargo from the hinterland. In the absence of significant gateway traffic, the port would have to necessarily depend on transshipment traffic. **Exhibit 2** provides a map of the existing and proposed container handling ports in India.

The project was initially proposed to be implemented directly by Government of Kerala (GoK). The first global tender in 2003-04 issued by GoK did not succeed. In 2004, GoK established the Vizhinjam International Seaport Limited (VISL), a company fully owned by GoK, as the implementing agency for the project in PPP mode using the Landlord model, with the private investor taking responsibility for all construction and financing including the basic infrastructure.

In 2007, two bids were received - one from a consortium consisting of Kaidi Electric Power Co. Ltd, China and Zoom Developers, Mumbai and the second from Gammon India Limited, Mumbai. The consortium led by Kaidi Electric Power Co. Ltd. was selected but Government of India (GOI) did not provide clearance for the project to be awarded to a Chinese consortium. In the second bid during 2008-09 M/s LANCO Infratech was the selected bidder. However, Zoom Developers (with a different consortium partner), whose bid was rejected, challenged the selection of LANCO in court. While the legal proceedings were going on LANCO decided to withdraw and the entire bidding was cancelled by the government. In the subsequent bid in 2010-12, M/s Welspun-Leighton Consortium was the single qualified bidder. However, it demanded a grant and sought some other benefits which were not acceptable to the government.

IFC STRATEGIC OPTIONS REPORT²

In November 2009, the Government hired IFC to help organize a bid transaction for private participation in the project. As a first step, a market assessment was undertaken with help of a team led by Drewry Shipping's international expert from UK. The market assessment revealed that in the changed economic environment after 2008-09, the projections of traffic at Vizhinjam were significantly lower than that indicated in the previous round of bidding. In a business as usual scenario, Vizhinjam Port was expected to attract low volume of gateway traffic, i.e. port traffic originating in or destined for Kerala. This was primarily because of low industrial activity in the immediate hinterland of the port. In the absence of significant gateway traffic, the port would have to necessarily depend on transshipment traffic.

Transshipment Traffic

According to the report, a port based primarily on transshipment traffic was a high-risk investment and unlikely to attract many credible and experienced investors. Large investments in greenfield ports were rarely planned based primarily on transshipment traffic

² Retrieved from http://www.vizhinjamport.in/downloads/Kerala_Port_SOR_FINAL.pdf

since transshipment traffic was unpredictable and shipping lines were known to switch from one port to another to gain even small advantages. Traffic risk could be mitigated by locking in a major shipping line as an investor. However, in the wake of substantial losses in 2008-09 and the continued uncertainty in the global trade scenario, very few major container shipping lines would have the appetite to bid for and invest in a greenfield port. Moreover, if a shipping line had plans to invest in the project, with a view to making it its transshipment hub, then competing shipping lines were likely to stay away, fearing a differential service and pricing. A port based primarily on transshipment traffic would also not have significant linkages and synergies with the local economy. As a result, one of the key priorities of the development of the state, was unlikely to be achieved.

Therefore, the report proposed that:

The Vizhinjam Project should be planned, not merely as a stand-alone port project, but as a larger development, whereby area near the port is developed as industrial and/or logistical hub, offering opportunity for synergies between the port and the Kerala economy and at the same time providing medium to long term gateway traffic growth at the port.

However, the establishment of an industrial/logistical hub near the port would generate traffic to the port only in the long term. In the short to medium term, the port would still have to depend on transshipment traffic. The Vizhinjam port would have to compete vigorously with ports such as Colombo and Vallarpadam which had a head start in terms of established operations, presence of global players in port operations, relationships with shipping lines and shippers and better logistical networks. International consultants hired by IFC (Drewry and Royal Haskoning) had simulated such a market assessment and concept planning for the port. Based on these studies and bidder consultations, "it is apparent that the Vizhinjam Port would have to adopt proactive strategies to attract transshipment cargo".

The project structure would have to set up the right incentives and checks and balances to ensure that design and construction of civil works (and terminals) would enable efficient port operations and investments. Specifically, the design of the port civil works (breakwater, quay walls, dredging and reclamation) should be such that the port catered to the largest container ships of 12,500 TEU capacity. Further, the design principles of the terminal facilities should support highly efficient operations. One requirement of this principle was that the port capacity should be adequate to ensure berth occupancy rates of not more than 55% to 58%, thereby ensuring a quick turnaround for ships calling at the port. Transshipment traffic in particular required disruption free operations at the port.

As such, strategies have to be developed to counter the general perception of labour related issues in Kerala. Some of the bidders consulted have suggested that establishment of a Special Economic Zone at the port may give comfort in this context to bidders and also the shipping lines that are potential customers for the transshipment traffic at the port.

Strategic tie-ups for cargo and traffic development

The management and operation of the port would have to be highly market oriented. This implied that strategic tie-ups were equally, if not more important to the success of the port than just constructing a "world-class" facility. While construction could be taken up through

public funded contracts, this implementation strategy had the risk of creating a white elephant with poorer economic and financial results in the medium to long term.

The report pointed to the need for involving the private parties in the project development from an early stage

However, it will be important to involve private investors/operator in the planning, design and construction of the port civil works and to give him independence in deciding on the nature and timing of terminal capacity addition in subsequent phases of the Project. The project structure has to set up the right incentives and checks and balances to ensure that design and construction of civil works (and terminals) is optimal from view point of efficient port operations and timing and extent of investments.

As such, the transaction structure would have to enable experienced private players to develop strategic tie-ups with port operators, shipping lines, shippers and other key players in the logistics value chain at an early stage and have a say in the planning, design and construction of the port.

Pricing of services

Since transshipment traffic was highly price sensitive, Drewry had suggested that the project would have to initially offer substantial discounts to the transshipment tariff offered at Colombo port and match the tariff at Tuticorin port for bulk cargo and vessel related charges. Once the port established efficient operations and was successful in forming strategic relations with shipping lines, it could increase tariffs in the medium term. It would be necessary for the bidders to have complete freedom to fix cargo and vessel related charges.

Financing of the project

Large investments in port infrastructure would be needed to construct a world class port with meaningful spinoffs for the Kerala economy. However, because of a highly competitive scenario, the port was projected to have relatively low revenues because of low traffic and low tariffs. The report mentioned:

Financial analysis indicates that the project has low debt capacity, i.e. the cash flows are not enough to support high levels of debt. Consequently, the project will need a large infusion of Equity. However, the return to equity investors is much lower than threshold returns required by investors. The project will be unviable without Government financial support.

International experience also indicated that there were very few greenfield projects where costs of breakwater, dredging, reclamation and quay walls had been fully financed and developed by a private investor. In cases where this could be done, the revenue potential was very high because their hinterland had high industrial growth, mines or for power plants with captive cargo and an ability to charge higher tariff because of relative lack of competition. In the case of Vizhinjam, several of these pre-conditions did not apply.

Since, the life of the port assets generally ranged from 50 to 100 years, depreciating them in the typical concession period of 25 to 40 years was not feasible.

Moreover, in India, commercial debt is generally not available beyond 18 years' tenure. Generally, investors are averse to projects that require long waiting periods to earn returns. In the case of Vizhinjam, traffic and tariff projections indicate that threshold returns required by investors might never be achieved if investors have to fully finance breakwater, reclamation and quay walls. Consultation with potential bidders has further confirmed this, as most parties consulted have indicated their inability to finance civil works.

Therefore, substantial financial support from the Government would be required for the project to make it financially viable. Such support was required primarily to meet the capital expenditure for breakwaters, dredging and reclamation and quay walls, apart from the external infrastructure such as road and rail linkages.

The quantum of financial support should be discovered through the bidding process, as the bid criterion. The bid technical parameters and process should be so defined that bidders had sufficient flexibility in adopting innovative business plans. At the same time, to conserve bid integrity and comparability a basic minimum technical specification should be outlined in the Request for Proposals.

Overall, the level of response to the bid would depend on a coherent and credible competitive strategy being conveyed to bidders through the various elements of the project structure. These elements included - construction of world class port infrastructure, role for investor in design and implementation of civil works, flexibility in deciding on timing and nature of capacity addition in later phases, commitment to establish a Special Economic Zone and freedom to set tariffs.

Landlord Port model

The IFC recommended the "Modified Landlord Model" for private participation. The landlord port was the dominant port model in larger and medium sized ports across the world. Under this model, the port authority acted as regulatory body and as landlord, while port operations were carried out by private companies.

The main features of the model were:

- a) GoK would be responsible for:
 - i. Developing a long-term master plan for the Port.
 - ii. Construction, operations and maintenance of the civil infrastructure, through an EPC contract awarded to a third party after seeking design inputs from the Concessionaire.
 - iii. Providing marine infrastructure at the Port and marine services to Users.
 - iv. Appointing the appropriate governmental agency to act as the conservator of the Port under the Indian Ports Act. The role of such Conservator was to oversee and ensure compliance with the Indian Ports Act and the Port By-Laws so that public health and maritime safety were maintained. The Conservator would also exercise police powers at the Port (e.g., power to board vessels; try ships, masters or the Concessionaire for offences committed; power of distraint over vessels and recovery of costs for such distraint; take actions to maintain defence and security of the port; take actions in emergencies).

- b) GoK would grant a concession on a BTO (Build to Order) basis to a qualified private firm/consortium for developing, operating, maintaining the suprastructure (terminal) and to use the civil infrastructure and suprastructure to provide cargo handling and other services to Users at the Port. The Concessionaire would be required to provide these services to the Users at specified performance levels.
- c) GoK would own the civil infrastructure and lease its possession and use (including rights to use existing or reclaimed land and waterfront) at nominal rates, to the Concessionaire for the purpose of implementing the Project.
- d) GoK could earn revenue from the Project by setting, levying and collecting port dues and other charges relating to the use of the civil infrastructure by the users. In addition, GoK would be entitled to set, levy and collect tariff from users for the provision of marine services. The Concessionaire would be entitled to set, levy and collect tariff from users for the use of suprastructure and the provision of cargo handling and other services.
- e) Although Project assets would be owned by the GoK, the Concessionaire would be entitled to create security over its leasehold rights in assets, its concession rights, the project assets and project revenues earned by the Concessionaire.
- f) If the Landlord Model was used, then bidders would be required to bid for the extent of government financial support required from GoK for developing and operating the supra-structure (terminal). Alternatively, in case bidders believed the construction and operation of the suprastructure (terminals) to be very profitable, they would be free to bid for a revenue share with GoK.

The Modified Landlord Model would essentially be the same as the Landlord Model, except in the following significant ways:

- a) Instead of GoK awarding the EPC contract for civil infrastructure to a third party, GoK would award the EPC contract as well as the concession for the suprastructure (terminals) to the same winning bidder.
- b) Each bidder would be required to simultaneously bid for an EPC contract for construction and development of the civil infrastructure and for the BTO concession for the development, construction, operations, management and maintenance of the suprastructure (terminals) and the provision of cargo handling services, marine services and other ancillary services to the users.
- c) If the Modified Landlord Model was used, then the bidders would be required to bid for: (1) the lump-sum EPC contract price; and (2) extent of government financial support required from the GoK for developing and operating the suprastructure, or alternatively, the concession fee that the bidder was willing to pay to the GoK. The aggregate financial impact for the GoK, in NPV terms, would be compared and the bid that had the least financial impact would be declared as the winning bid.

VIZHINJAM PORT MODEL

In November 2013, the GoK approved a Design, Build, Finance, Operate and Transfer (DBFOT) PPP model for development of the project. The project structure was based on the Landlord Port Model and the role of GoK/VISL and Concessionaire was as follows:

GoK would grant a concession on a DBFOT basis for developing, operating, maintaining and managing the entire Port including the civil infrastructure and supra-structure (terminal) and to provide cargo handling services and other ancillary services to users at the port. However, the funding of pre identified civil infrastructure works (breakwater and fishing harbor) would

be borne by GoK/VISL. The Concessionaire would be required to undertake to provide these services to the users at specified performance levels.

Concessionaire would be entitled to set, levy and collect tariff from users for the use of all infrastructure at the Port and the provision of cargo handling services and other ancillary services to users. GoK would notify the tariff set for vessel and container handling related services in accordance with the concession agreement, which would be the ceiling. The Concessionaire would also be entitled to generate revenues from the port estate development for the entire concession period.

Project assets would be owned by GoK, the Concessionaire would be entitled to create security over its leasehold rights in assets, its concession rights, the project assets and project revenues earned by the Concessionaire. These assets would be transferred to the GoK on expiration or early termination of the concession.

During the bid process, the bidders would be required to bid for the extent of government financial support required, in form of capital grant as Viability Gap Funding³, from the GoI/GoK. **Exhibit 3** provides details of eligibility condition for viability gap funding.

Capital investment

The capital investment would be divided into two parts - Funded Works and PPP Works. The provision of Funded Works had been introduced by the Planning Commission in the Model Concession Agreement (MCA) for the work that was supposed to be undertaken by the government using an EPC contract. As Funded Works, these works would be executed by the same Concessionaire on the basis of reimbursement of specified costs as a lump sum amount. This was introduced in the MCA with the aim of avoiding interface disputes on engagement of different agencies for work of same project, thereby reducing conflicts, time/cost over runs and ensuring timely completion of the project. The concession terms were modified to include construction of breakwater and fishing harbour as Funded Works to be funded by the State.

Funded Works would involve the construction of:

- Breakwater: approximately 3,100m of breakwater to protect the harbour basin from waves and swells. The breakwater would facilitate safe navigation conditions into and around the port providing a haven for port vessels during extreme ocean conditions thereby creating an acceptable wave climate for the design of port infrastructure.
- Fishery harbour and associated onshore facilities: During the initial community consultation fishing communities had expressed apprehensions that the port would adversely affect their livelihood. The existing Vizhinjam fishing harbour had berthing facility of 500m and was being utilized to its peak causing congestion in the utilization of the harbour. In order to ease the congestion in the existing fishing harbor and provide additional facilities for the local population, VISL had proposed to provide fishery berths along the sea side of north breakwater of the proposed Vizhinjam port “as part of Corporate Social Responsibility”.

³ Viability Gap Funding (VGF) is designed to provide capital support to PPP projects which would not otherwise be financially viable. VGF has the effect of reducing the revenue required to recover costs and provide a financially attractive return for the private partner.

The lump sum costs of all construction works comprising Funded Works was INR 1,463 crore⁴. This amount would be modified to the extent of variation in the Wholesale Price Index (WPI) occurring between the Bid Date and the Appointed Date. Any expenditure in excess, except on account of Force Majeure, Change of Scope or Change in Law, as the case may, would be borne entirely by the Concessionaire

The PPP Works would comprise:

- Approximately 800m of quay for container loading/unloading operation, basin area for port crafts⁵, and 500m fish landing berth;
- Dredging to create harbour basin and approach channel. Dredging works are required to lower the seabed for safe navigation and berthing of vessels. The proposed port site was characterized by naturally available deep-water depths with 20m contour located at a distance of less than 800m from the shore. This would substantially reduce the dredging cost, enabling the port to provide berthing ability for the largest container vessels (up to 18,000 TEU).
- Reclamation with the dredged material to create onshore facilities and other associated civil works. The new port of Vizhinjam would be constructed on reclaimed land in front of the existing coastline south of the existing fishery port. Reclamation works would be required at the back side of the berth to create land for building container storage yard and spaces for other utilities required for the port. The port would utilize reclaimed land for most of the onshore port facilities. The dredging operations were expected to meet most of the demand for reclamation material.
- The proposed facility was the Phase 1 development with the intention to expand the facility in future. All design and construction works related to the Phase 1 development would take into account future development planned by VISL.

Nature of work	Funding pattern
PPP Work: Dredging and reclamation, development of berths, roads, substations, superstructure, equipment and operation INR 4,089 crore	Upto INR 1,635 crore through VGF, equally by Government of India and GoK. The Grant would be disbursed to the Concessionaire by way of Equity Support and as O&M Support. Balance funding by the Concessionaire The project was to be implemented in four phases with a rated capacity of 6 lakh (0.6 million) TEUs on commercial operation date (COD), to be enhanced to one million TEUs within 10 years of COD. The port shall be capable of accommodating vessels with capacity up to 18,500 TEUs. The Concession period would be 40 years commencing from the Appointed Date which shall be extendable for further 20 years at the option of the Concessionaire subject to capacity augmentation to three million TEUs by 30 th year of the concession period.
Funded Work	The construction of 3.1 kilometer (km) long breakwater and a new fishing harbour would be done by the

⁴ 1 crore = 10 million

⁵ This includes tug boats for berthing, stopping and turning maneuvers for container and other vessels. Other port crafts include mooring launch and pilot cum survey launch.

INR 1,463 crore	Concessionaire as “funded works” for which the GoK would finance the entire amount of INR 1,463 crore.
Cost of external infrastructure INR 1,973 crore	Entire funding by GoK.
Total cost INR 7,525 crore	

All project assets would be owned by the GoK, and the Concessionaire would be entitled to create security over its leasehold rights in assets, its concession rights, the Project assets and Project revenues earned by the Concessionaire. These assets would be transferred to GoK on expiration or early termination of the concession.

FINANCIAL TERMS OF THE CONCESSION AGREEMENT⁶

According to the feasibility study the key elements of financial viability were the port traffic, concession period and port tariffs. The concession period should be long enough to enable the recovery of capital costs with adequate returns to debt and equity investors. The concession period would have to be higher, given capital costs, if traffic and/or tariffs are lower because of competition or macroeconomic factors.

Given below are key financial details of the concession agreement:

Concession fee

Starting from the fifteenth year after COD (Commercial Operation Date) the Concessionaire would pay a concession fee as a percentage of the Realizable Fee during the particular year. The percentage would start at 1% during the fifteenth year and increase by an additional 1% each year, subject to a ceiling of 40%.

In the first year following the COD, the Realisable Fee would be the higher of (a) the actual Fee realised by the Concessionaire and (b) INR 100 crore (Rupees one hundred crore).

The Realisable Fee for computing the Concession Fee for each subsequent year would be deemed to be the higher of (a) the actual Fee and (b) 3% greater than the Realisable Fee in the immediately preceding year.

Collection and appropriation of User Fee

The Concessionaire would set, levy and collect tariff for the use of all infrastructure at the Port, and the provision of cargo handling services and other ancillary services in accordance with the fee Schedule in the concession agreement referred to as the Base Fee. The base fee would represent the maximum fee and the concessionaire could charge lower fees. The Concessionaire could, in its discretion, revise the Fee annually to reflect the variation in Price Index or in accordance with the rates prevailing at other Major Ports, as the case may be. The fees would be subject to upward revision for any unforeseen event including any change in scope or other modifications in the Port that were not contemplated and which required a capital expenditure of INR 50 crore (Rupees fifty crore) or more.

⁶ Retrieved from <http://www.vizhinjamport.in/downloads/Executed-Concession-Agreement-Development-Vizhinjam-International-Seaport-17-Aug-15.pdf>

The fee schedule could be kept in abeyance if the Port Authority was satisfied that adequate capacity of port infrastructure was available and the user charges were being determined by free competition. At the end of 10 years from the “Appointed Date”, the user charges would be determined entirely by competition and the Fee Schedule would remain in abeyance thereafter.

For any goods or services not specified in the Fee Schedule, the Concessionaire would specify a fee not exceeding the rates prevalent from time to time in other similar ports in India and Asia.

Revenue Shortfall loan

A revenue shortfall would occur if the “Realisable Fee” in any Accounting Year fell short of the “Subsistence Revenue” as a result of an Indirect Political Event, a Political Event or an Authority Default.

“Realisable Fee” meant all the Fee due and realisable under this agreement, with or without any discounts or reduction in Fee, but did not include fees that the Concessionaire had not been able to realise after due diligence and best efforts. The amount would be determined on the basis of its provisional accounts or the audited accounts, as the case may be.

“Subsistence Revenue” meant the total amount of Fee revenue that was required by the Concessionaire in an Accounting Year to meet the sum of:

- (a) O&M Expenses, subject to an annual ceiling of 3% (three percent) of the Total Project Cost, plus Grant, if any, during the first Accounting Year after COD, to be revised for each subsequent year to reflect the variations in Price Index occurring between COD and commencement of such Accounting Year, and
- (b) Debt Service in such Accounting Year

In the event of the revenue shortfall, the Port Authority would provide a loan for meeting such shortfall at an interest rate equal to 2% above the bank rate.

Effect of variations in traffic growth

In the event that the Actual Average Traffic fell short of, or exceeded, the Target Traffic by more than 5%, the Concession Period would be modified. Traffic equivalent to 60% of the capacity of the port, estimated to be 600,000 TEUs per annum, would be treated as the “**Target Traffic**”.

For every 2% shortfall, the Concession Period would be increased by one year, subject to a maximum of 10 years; for every 2% excess as compared to the Target Traffic, the Concession Period would be reduced by 6 months subject to a maximum of 3 years. In lieu of a reduction in Concession Period the Concessionaire could elect to pay, in addition to the Concession Fee that would be due and payable if the Concession Period were not reduced, a further premium equal to 10% of the Realisable Fee in the respective years.

Development of Port Estate

The Concessionaire could utilize 30% of the land acquired for the project by GoK for “Port estate Development” which would include residential and commercial buildings and space.

The Concessionaire would be able to generate revenues from the Port Estate Development for the entire concession period. The Concessionaire would undertake Port Estate Development at its cost on the land specified for commercial purposes with the right to sublicense any or all parts by means of Project Agreements. All revenues accruing from Port Estate Development would be included in Other Business undertaken by the Concessionaire for the optimum utilisation of the Project Assets.

Starting from the seventh year after the start of commercial operations, the Concessionaire would pay to the Authority 10% of the gross revenues received from the Other Business in each month (the "Revenue Share from Other Business"). "Provided, however, that in the event of any sale of goods or commodities, or provision of any services by the Concessionaire, only 10% (ten percent) of the proceeds thereof shall be deemed to be gross revenues of the Concessionaire, and revenue share on such gross revenues shall be payable by the Concessionaire."

FEASIBILITY STUDY⁷

Ernst & Young, the Financial Consultant and Transaction Adviser, prepared and submitted a Feasibility Report, which was approved by the government in May 2014. The feasibility Report was based on the following assumptions.

Concession & Construction Period

The Concession Period considered was 40 years, inclusive of 4 years of construction period for port development and 6 years for Port Estate Development. Revenues would commence from the sixth year of the appointed date. Only Phase-1 Port development was envisaged to be constructed in this period as per the draft Concession Agreement. Project construction would start in 2014-15 and be completed in 2017-18 for Port Development and 2019-20 for Port Estate Development. Phase - I mandatory capacity was 1.0 million TEU.

A year on year escalation on the base cost was assumed at 5% p.a., as per Planning Commission guidelines for various infrastructure projects. An escalation of 25% was added to the total hard cost which primarily included provision for interest during construction and other contingencies. The estimated total project cost (TPC) of the project was pegged at INR 4,089 crore, excluding the cost of funded works. Details are provided in **Exhibit 4**.

Traffic forecast

Colombo port was the primary competitor for Vizhinjam port for transshipment market and the current market dynamics suggested that only the Indian Sub-continent (ISC) transshipment traffic market could be catered to by the port. Container traffic was forecasted to grow at 8% per annum. Gateway traffic, loaded and empty, was forecasted to grow at 10% & 8% respectively. The port was expected to have transshipment and gateway traffic in the ratio of approximately 75%-25%.

Based on the port components planned for Phase-1 development, the minimum container terminal capacity for handling the projected traffic was 1,000,000 TEU. This terminal handling capacity would be exhausted in the FY 2026 with forecasted traffic of 951,467 TEU. However, with the state-of-art equipment to handle the container traffic the port could achieve a

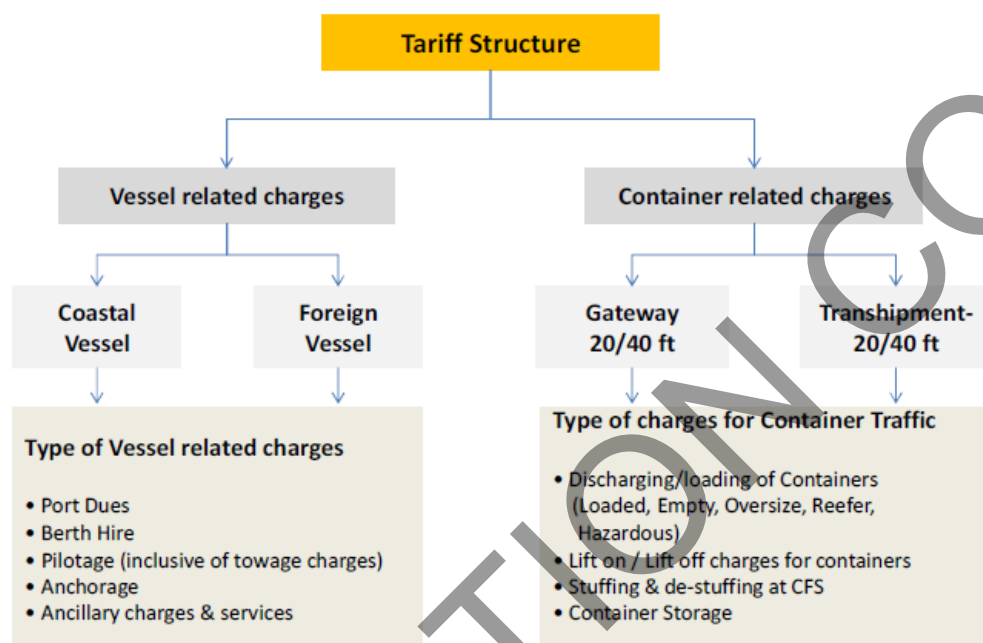
⁷ Retrieved from <http://www.vizhinjamport.in/download/Feasibility-Report.pdf>

maximum utilization rate of about 125%. Therefore, container terminal capacity in Phase-1 development of Vizhinjam Port was expected to cater to a maximum traffic of 1,250,000 TEU which would be achieved in FY 2030 and remain constant for the rest of the horizon period as capacity augmentation was not envisaged in Phase-1 Port Development.

Details of traffic forecast are available in **Exhibit 5**

TARIFF STRUCTURE

The diagram below shows the tariff structure for ports.



Tariffs for Vizhinjam were benchmarked to the charges of other major container handling ports in the region. Ports situated within 750 km from Vizhinjam had been studied for both vessel and container tariffs. The cargo tariff structure study was limited to container cargo tariff since the Phase-1 port development did not envisage any bulk cargo berth terminal capacity. The table below shows the main competitors for vessel and container related charges

Vessel Related Charges		Container Related Charges	
Coastal Vessel	Foreign Vessel	Gateway	Transshipment
Cochin, Chennai, Ennore, New Mangalore & Tuticorin	Cochin, Chennai, Ennore, New Mangalore, Tuticorin, Colombo, Dubai & Salalah	Cochin, Chennai & Tuticorin	Colombo, Cochin, Dubai & Salalah

In order to be competitive, and generate initial traffic base, the proposed Vizhinjam port would offer a substantial discount from competition. After the initial build-up of traffic, the discount margin could be reduced. According to the feasibility study, foreign ports like Colombo offered discounts (15-20%) over the published rates depending on the frequency of the calls and throughput generated by the shipping line. The financial analysis used a discount of 35%. The tariff rate would be increased by the WPI increase, assumed to be 7% per year over the forecast period.

Port Revenue and operating costs forecasts are provided in **Exhibit 6**.

Port estate Development

As per the concession agreement a maximum of 30% of the land acquired for development of the project would be utilized for commercial exploitation. Commercial Port estate projects generally comprise of retail mall space, Grade A and Grade B office spaces, different categories of hotels and service apartments, residential complexes etc. The table below gives the built-up area distribution of port estate components.

Project Components	Area Distribution (Sq ft)	Share of development (%)
Residential	4,573,800	33%
Retail	3,841,992	42%
Commercial	1,372,140	15%
Mid-market Hotel	182,952	4%
Luxury Hotel	259,182	6%

Construction phasing formed one of the key assumptions in any Port estate project. The decision on phasing depended mainly on the demand-supply scenario in each of the type of projects listed above. The phasing would have to take into account the financial viability by estimating the expected dynamics between absorption rates and capital/rental values. A six-year construction phasing had been proposed keeping in view the large area of 105 acres available for port estate development and the overall growth in the port traffic. Majority of the construction activity had been proposed from fourth year onwards and the revenues would start to accrue from the project from sixth year from the appointed date.

Revenue estimates were based on primary and secondary survey of the current rental rates prevailing in the market. Revenue streams for city side development had been considered for retail, commercial and hospitality sectors. The hotel industry formed a major portion of the revenue from city side development. The revenue for city side development in Vizhinjam had been estimated till the year 2054, i.e. for 40-year concession period.

Exhibit 7 provides operational assumptions and capital expenditure for Port Estate Development. **Exhibit 8** provides the projected revenue and operating cost estimates for Port Estate Development.

FINANCIAL ANALYSIS

In order to evaluate the financial viability of the Project and the amount of viability gap funding required the following scenarios were explored:

Base case scenario: No viability gap funding or Port Estate Development

Scenario 1: Government support as Viability Gap Funding 60% of the Total Project Cost offered

Scenario 2: Viability Gap Funding 40.7% of the Total Project Cost offered along with land for commercial exploitation

The Viability Gap Funding in the two cases was chosen so as to provide a 15% Equity IRR.

The Weighted Average Cost of Capital was assumed at 10.6%.

The cash flow forecasts for the base case and the two scenarios are presented in the accompanying workbook. The analysis generated the following profitability metrics.

Key Profitability Metrics	Base Case	Scenario 1	Scenario 2
Project IRR	8.70%	8.75%	10.57%
Equity IRR	11.54%	15.00%	15.00%
NPV Revenue share	INR 276		
VGf (in %)		60.00%	40.70%
VGf (in INR crore)		INR 2,454 crores	INR 1,664 crores
Total Project Cost (in INR crore)	INR 4,089 crores	INR 4,089 crores	INR 4,089 crores

On the basis of the financial analysis the report concluded that the project was not financially viable and would have to be supplemented with a mix of viability gap funding and additional revenues from Port estate development.

ADANI PORTS AND SEZ LTD (APSL)

VISL invited (December 2013) International Competitive Bids (ICB)/Global Tenders for selection of PPP Concessionaire. According to the terms of RFP, selection of bidder was to be based on the highest premium offered to GoK or lowest grant demanded from GoK. Maximum grant that can be demanded by way of Viability Gap Funding (VGf) was capped at INR 1,635 crore, being 40 percent of the TPC. Adani Ports and SEZ Ltd., a subsidiary of Adani Enterprises Limited, was contemplating a bid for the project.

APSL was a subsidiary of Adani Enterprises Limited, the flagship company of the Adani Group. The Adani Group had significant interests across resources (coal mining and trading), logistics (ports and logistics, shipping and rail), energy (power generation and transmission) and other ancillary industries, with a presence in India, Indonesia, Singapore, Australia, China and the Middle East. The Adani Group included three listed companies in India, Adani Enterprises Limited, Adani Power Limited and APSL. APSL is one of India's largest private developers and operators of ports and related infrastructure and provides fully integrated port and logistics services.

APSL developed and operated the port at Mundra, Gujarat, the largest non-major port in India by volume, handling 82 mmt for the fiscal year 2013. The Mundra Port could accommodate capesize container cargo vessels of up to 10,000 TEUs, and VLCCs and ULCCs of up to 360,000 DWTs. It provided infrastructure, leasing and logistics services at the Mundra Port and through its surrounding infrastructure. This included a special economic zone in the surrounding area (the "Mundra SEZ"), one of the largest operating port-based multi-product special economic zones in India. At the Mundra Port, APSL had developed and operated three container terminals. It provided marine, intra-port transport, storage and handling and other services at Container Terminals 1, 2 and 3 and an automobile handling facility. It had entered into a sub-concession agreement with Mundra International Container Terminal Private Limited (MICTL) to operate Container Terminal 1 and planned to enter into a sub-concession agreement with Adani International Container Terminal Limited (AICTL) to operate Container Terminal 3. It operated Container Terminal 2 and the automobile handling facility

by itself. Total Revenue attributable to the Mundra Port accounted for 92.7%, 89.9% and 85.2% of the total revenue for the fiscal years 2011, 2012 and 2013, respectively.

APSL operated three complementary terminals at two ports: a container terminal and a multi-cargo terminal at the Hazira Port and a dry bulk cargo terminal at the Dahej Port. These terminals functioned as “twin” ports, allowing APSL to service a variety of cargo in the central and southwestern hinterlands of India. APSL was also developing coal handling terminals at the port at Mormugao, Goa, and at the port at Visakhapatnam, Andhra Pradesh (also known as the “Vizag Port”) as well as a bulk cargo terminal at the port at Kandla, Gujarat.

APSL had to decide whether to bid for the project and on what terms.

INSPECTION COPY

Exhibit 1

Vizhinjam port location with respect to International East-West Shipping Route

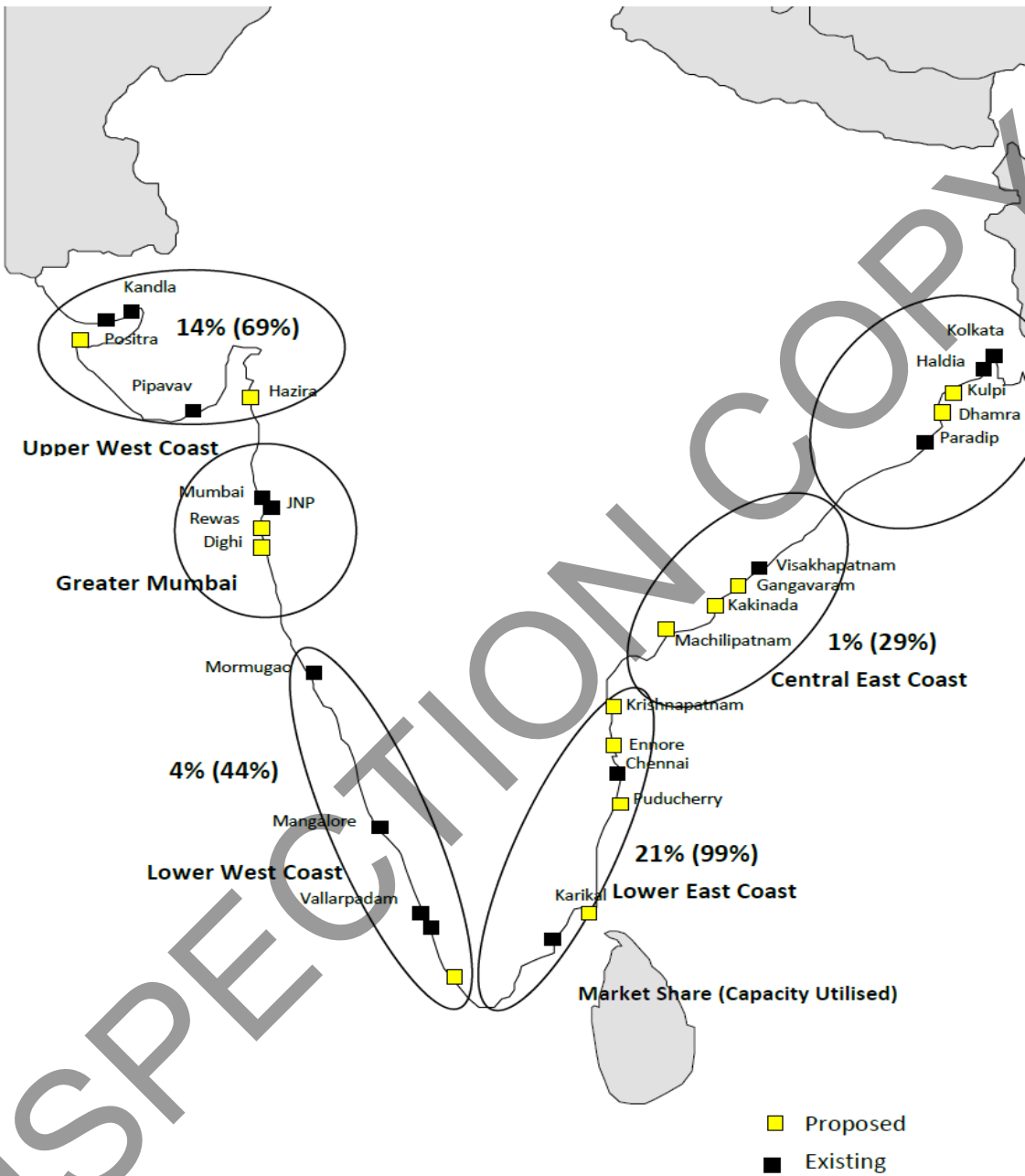


Figure 1.2 Vizhinjam Port Location with respect to International East-West Shipping Route

Source: Development of Vizhinjam Port, Detailed Project Report May 2013

Exhibit 2

Existing and proposed container handling ports in India



Source: Development of Vizhinjam Port, Detailed Project Report May 2013

Exhibit 3

Guidelines for forwarding proposals to DEA to ascertain their eligibility under the Viability Gap Funding Scheme, Memorandum to the Department of Economic Affairs

A. Conditions of the VGF Scheme

1.	Whether the project proposal has been posed by a Government or statutory entity which owns the underlying assets
2.	Whether the project is to be implemented, i.e. developed, financed, constructed, maintained and operated for the Project Term by a Private Sector Company
3.	Is the project from the sectors identified in the guidelines
4.	Whether the Private Sector Company will be selected by the Government or a statutory entity that owns the project through a transparent and open competitive bidding process
5.	Whether the Project provides a service against payment of a pre-determined tariff or user charge
6.	Whether user charges/tariff has been fixed by Government or a statutory authority
7.	Whether the Government/statutory entity making the proposal has certified/will be able to certify with reasons; a) That the tariff/user charge cannot be increased to eliminate or reduce the viability gap of the PPP; b) That the Project Term cannot be increased for reducing the viability gap; c) That the capital costs are reasonable and based on the standards and specifications normally applicable to such projects and that the capital costs cannot be further restricted for reducing the viability gap
8.	Is the total VGF within the gap stipulated in the guidelines
9.	Whether the proposed project is (or will be) based on standardised/model documents duly approved by the respective Government

B. Other project related information (if available)

1.	Total Project Cost (in INR Cr.)
2.	VGF sought from GoI (in INR Cr.)
3.	VGF as a percent of Total project cost
4.	Additional grant from the Sponsoring Authority
5.	Construction period (from Financial closure)
6.	Likely year in which VGF sought
7.	Is the project viable without VGF
8.	If not, is it viable with VGF
9.	Status of the concession agreement • Has it been finalized • If not, is it proposed to be based on a model document

Source: Scheme and Guidelines for Financial Support to Public Private Partnerships in Infrastructure, 2013, Government of India, Ministry of Finance, Department of Economic Affairs

Exhibit 4
Project Capital Expenditure

Financial year	Total project cost (in INR Cr)
Civil costs	
Project preliminaries and Site development	11
Dredging & reclamation	720
Berths	426
Buildings	28
Container yard	218
Equipment	
RMQC	723
Reach stackers	7
RTG (Yard)	270
Empty container handler	17
ITVs	46
Maintenance vehicles	0
Workshop equipment	6
Spares	54
Utilities and Others	110
Port crafts and Aids to navigation	360
Gates complex & Road development	120
Engineering and project management fee at 5%	156
Total Base (Civil and Equipment) Cost	3,272
Escalation for Interest during construction and other financing cost @ 25% as per Planning Commission	818
Total Project Cost	4,089

Source: Ernst & Young Feasibility Report (2014), Table 9-2: Phase-1 Port Development Cost Estimate Summary

Base	2015-2025	2026-2036	2037-2047	2048-2054
5.5%	5.8%	6.1%	6.4%	6.7%

Source: Ernst & Young Feasibility report (2014), Table 9-4: Capex cost Escalation for Port Development

Exhibit 5

Traffic Projections

Projected Container Traffic ('000 TEU) for Vizhinjam Port 2018-2054 (for both Phase 1 and 2)

Traffic ('000 TEUs)	2019	2023	2028	2033	2038	2043	2048	2054
Gateway-Loaded	26	92	160	266	380	493	615	778
Gateway-Empty	11	40	53	898	94	123	154	196
Gateway-Total	37	132	213	354	474	616	769	973
Transshipment	112	566	878	1,213	1,550	1,817	2,055	2,314
Total	149	698	1,091	1,567	2,204	2,433	2,824	3,287

Source: Ernst & Young (2014) Feasibility Report, Table 6-1: Projected Container Traffic ('000 TEU) for Vizhinjam Port 2018-2054 (for both Phase 1 and 2)

Traffic based on Phase 1 investment only

Traffic ('000 TEUs)	2019	2023	2028	2033	2038	2043	2048	2054
Gateway-Loaded	26	92	160	194	194	194	194	194
Gateway-Empty	11	40	53	64	64	64	64	64
Gateway-Total	37	132	213	258	258	258	258	258
Transshipment	112	566	878	992	992	992	992	992
Total	149	698	1,091	1,250	1,250	1,250	1,250	1,250

Source: Ernst & Young Feasibility Report (2014), Page 14

Terminal Development Capacity Summary for Phase 1 Development

Phase-1: Minimum Total Container Capacity Planned (TEU)	1,000,000	1,000,000
Phase-1: Container Traffic Forecast (TEU)	951,467	1,250,025
Capacity Utilization	100%	125%
Planned terminal capacity achieved in the year	2026	2030

Source: Ernst & Young Feasibility report (2014), Table 6-2: Terminal Development Capacity Summary for Phase 1 Development

Exhibit 6

Port revenues

Total Revenue Forecast for Vessel & Container related charges (INR crores)

	2019	2023	2028	2033	2038	2043	2048	2054
Vessel Charges	53	150	248	392	505	652	839	1141
Gateway	9	56	84	176	232	303	386	523
Transshipment	27	179	297	521	672	867	1118	1517
Miscellaneous*	4	27	44	82	106	138	177	240
Total (INR Crores)	93	412	673	1171	1515	1960	2520	3421

Source: Ernst & Young Feasibility Report (2014), Table 9-13: Total Revenue Forecast for Vessel & Container related charges

Revenue escalation assumptions

Base	2015	2026	2037	2048
	2025	2036	2047	2054
5.5%	5.8%	6.1%	6.4%	6.7%

Source: Ernst & Young Feasibility report (2014), Table 9-14: Revenue escalation assumptions

O&M Cost Summary for Port Development (INR crores)

Particulars	2019	2023	2028	2033	2038	2043	2048	2054
Power	1	4	7	13	17	22	30	42
Fuel	5	28	46	87	115	152	201	281
Repair and maintenance - Civil	19	23	31	41	54	71	95	133
Repair and maintenance - Equipment	30	37	49	65	86	114	152	212
Insurance	41	52	68	91	120	159	210	294
Labour cost	6	35	58	109	144	191	253	354
Salaries/Overhead/Admin and General Expenses	8	46	78	145	192	255	337	472
Total	110	225	337	551	728	964	1,278	1,788

Source: Ernst & Young Feasibility Report (2014), Table 9-21: O&M Cost Summary for Port Development

Exhibit 7

Operational assumptions and capital expenditure for Port Estate Development.

Operational assumptions

Particulars	Unit	Assumption
Total site area	Ha	142
Total site area	Acres	350
Land for port estate development	Acres	105
Other Assumptions		
Ground coverage - Residential	%	65%
Ground coverage - Retail/commercial	%	40%
Ground coverage - Hotels	%	40%
Start date of construction	Year	2015
Start date of operations	Year	2019
Start date of operations - Luxury hotel	Year	2020
Total construction period	No.	6
Hotel FAR	No.	1
Residential FAR	No.	3
Financing Assumptions		
Debt	%	50%
Equity	%	50%
Rate of interest	%	13%
No. of years for repayment	No.	5
Moratorium (yrs)	No.	4
Start year of disbursement	Year	2015
Start year of repayment	Year	2019

Source: Ernst & Young Feasibility Report (2014), Table 9-6: Assumptions for Port Estate Development

Capital expenditure

Particulars	Amount (in INR crore)	Percentage of total
Residential	1172.0	35%
Retail	1055.0	31%
Commercial	459.0	14%
Mid-market Hotel	81.0	2%
Luxury Hotel	180.0	5%
Total base cost	2947.0	88%
Preliminary expenses	116.0	3%
Contingencies	175.0	5%
Financing fees	11.2	0%
IDC	111.0	3%
Total Capex	3360.2	100%

Source: Ernst & Young Feasibility Report (2014), Table 9-7: CAPEX - Port estate Development Distribution

Exhibit 8

Projected revenue and operating cost estimates for Port Estate Development

Projected revenue

Particulars	2020	2023	2028	2033	2038	2043	2048	2054
Rentals								
Residential	29	51	101	165	209	264	331	431
Retail	140	245	488	800	1014	1277	1605	2087
Commercial	29	51	102	168	213	268	336	437
Total	199	347	692	1131	1437	1809	2272	2956
Hotels								
Mid - market	46	80	160	247	314	395	498	645
Luxury	57	100	199	307	389	490	617	800
Total	103	180	359	554	703	885	1114	1446
Grand total	302	527	1051	1687	2139	2693	3387	4401

Source: Ernst & Young Feasibility Report, Table 9-18: Projected revenue: Port Estate Development

Operating cost estimates

Particulars	2019	2023	2028	2033	2038	2043	2048	2054
Power	1	4	7	13	17	22	30	42
Fuel	5	28	46	87	115	152	201	281
Repair and maintenance - Civil	19	23	31	41	54	71	95	133
Repair and maintenance - Equipment	30	37	49	65	86	114	152	212
Insurance	41	52	68	91	120	159	210	294
Labour cost	6	35	58	109	144	191	253	354
Salaries/Overhead/Admin and General Expenses	8	46	78	145	192	255	337	472
Total	110	225	337	551	728	964	1,278	1,788

Source: Ernst & Young Feasibility Report, Table 9-21: O&M Cost Summary for Port Development